

1. Use the quadratic formula to solve $2x^2 + 7x - 15 = 0$. $a = 2, b = 7, c = -15$

$$x = \frac{-7 \pm \sqrt{49 + 120}}{4}$$

$$x = \frac{-7 \pm \sqrt{169}}{4}$$

$$x = \frac{-7 + 13}{4}$$

$$x = \frac{6}{4} = \frac{3}{2}$$

$$x = \frac{-7 - 13}{4}$$

$$x = \frac{-20}{4} = -5$$

2. Solve the equation, $3x^2 + 9x + 3 = -3$, by factoring.

$$(3x + 6)(x + 1) = 0$$

$$3x + 6 = 0$$

$$x = -2$$

$$x + 1 = 0$$

$$x = -1$$

3. Solve the equation, $x^2 = 12 - 4x$, by completing the square.

$$(x^2 + 4x + 4) - 12 - 4 = 0$$

$$(x + 2)^2 - 16 = 0$$

$$\sqrt{(x + 2)^2} = \sqrt{16}$$

$$x + 2 = \pm 4$$

$$x + 2 = 4$$

$$x = 2$$

$$x + 2 = -4$$

$$x = -6$$

4. Solve the equation, $3x^2 + 4 = 8x$, with the method of your choice.

$$3x^2 - 8x + 4 = 0$$

$$(3x - 2)(x - 2) = 0$$

$$3x - 2 = 0$$

$$x = \frac{2}{3}$$

$$x - 2 = 0$$

$$x = 2$$

The compact disc (CD) was invented in 1979. In 1982, CDs were made available to the public. CDs quickly became the most efficient way of storing music, replacing vinyl records and cassette tapes. The function $C(x) = -2x^2 + 38x + 40$ models the sales, in hundreds of millions of dollars, of compact discs in the years since 1990.

5. Solve $-2x^2 + 38x + 40 = 0$ by factoring.

$$-2(x^2 - 19x - 20) = 0$$

$$-16(x - 20)(x + 1) = 0$$

$$x - 20 = 0$$

$$x = 20$$

$$x + 1 = 0$$

$$x = -1$$

6. Interpret the solutions and determine if the solutions are viable.

The sales were \$0 in 1989 (-1 years from 1990) and 2010 ($+20$ years from 1990).

$x = -1$ is not a viable solution in the context of this problem since it was before 1990.

A statistician gathers data over several years, starting in 1995, on the average number of trips a household makes to run errands. The statistician uses the data to find the equation, $E(x) = -2x^2 + 68x + 480$, which models the average annual number of trips for errands per household for years since 1995.

7. Solve $-2x^2 + 68x + 480 = 0$ using the quadratic formula.

$$x = \frac{-68 \pm \sqrt{4,624 + 3,840}}{-4}$$

$$x = \frac{-68 \pm \sqrt{8,464}}{-4}$$

$$x = \frac{-68 \pm 92}{-4}$$

$$x = \frac{-68 + 92}{-4}$$

$$x = -6$$

$$x = \frac{-68 - 92}{-4}$$

$$x = 40$$

8. Interpret the solutions and determine if the solutions are viable.

It was 6 years before 1995 (1989), and 40 years after 1995 (2035), that the number of trips for errands was 0.

$x = 40$ is viable only if treated as a prediction. $x = -6$ is not viable because it does not fit the constraints of the problem.

A firework is launched from a hill that is 80 feet (ft) above a lake. The firework will descend into the lake after it has exploded at its maximum height. The firework's height, in ft, above the lake's surface is given by $h(t) = -16x^2 + 64x + 80$, where x is the time in seconds.

9. Solve $0 = -16x^2 + 64x + 80$ using the method of your choice.

$$-16(x^2 - 4x - 5) = 0$$

$$-16(x - 5)(x + 1) = 0$$

$$x - 5 = 0$$

$$x = 5$$

$$x + 1 = 0$$

$$x = -1$$

10. Interpret the solutions and determine if the solutions are viable.

After 5 seconds, the firework reaches the lake's surface.

$x = -1$ is not viable because you can't have a negative number of seconds.